

- 3 Cystic fibrosis is an autosomal recessive genetic disease. People with cystic fibrosis have a homozygous recessive genotype.

(a) Explain the meaning of the terms homozygous and recessive.

homozygous

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recessive

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[2]

(b) (i) In 2020:

- there were 10 800 people with cystic fibrosis in the UK
- the UK population was estimated to be 67 100 000 people.

A proportion of people in the UK population are heterozygous for the gene that causes cystic fibrosis and do **not** have symptoms of the disease.

Use the Hardy–Weinberg principle to calculate the number of people in the UK population who are expected to be heterozygous for the gene that causes cystic fibrosis.

The two equations for the Hardy–Weinberg principle are provided.

equation 1

$$p + q = 1$$

equation 2

$$p^2 + 2pq + q^2 = 1$$

p = frequency of the dominant allele

q = frequency of the recessive allele

p^2 = frequency of the homozygous dominant genotype

$2pq$ = frequency of the heterozygous genotype

q^2 = frequency of the homozygous recessive genotype

The first stage of the calculation has been completed for you.

$$q^2 = \frac{10\,800}{67\,100\,000}$$

$$q =$$

$$p =$$

$$2pq =$$

number of people in the UK expected to be heterozygous for the gene = [3]

- (ii) The Hardy–Weinberg principle provides a useful estimate of the number of people in the UK who are heterozygous for cystic fibrosis. However, the estimate is lower than the actual number. This underestimation occurs because **not** all the conditions of the Hardy–Weinberg principle apply.

In the UK in 2020, the mean life expectancy of:

- people with cystic fibrosis was approximately 50 years
- **all** people was approximately 80 years.

Explain how this information accounts for the underestimation of the number of people in the UK that are heterozygous for cystic fibrosis.

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- (c) A screening programme for cystic fibrosis was introduced in 2007 for all children born in the UK. Children are tested within seven days of their birth. Children identified from the screening programme as being at high risk of having cystic fibrosis can have a genetic test to confirm whether they have the disease.
- (i) Table 3.1 shows the median predicted life expectancy for people born in the UK who have cystic fibrosis. Predictions are shown for people born in 2008, 2012, 2016 and 2020.

Table 3.1

year of birth	median predicted life expectancy /years
2008	38.8
2012	43.5
2016	47.0
2020	50.6

Describe the trend shown in Table 3.1 **and** outline how early screening for cystic fibrosis may have contributed to this trend.

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- (ii) In many countries, a genetic test for cystic fibrosis is available to **adults** who do **not** have cystic fibrosis but have a family member who either has cystic fibrosis or is heterozygous for the gene that causes cystic fibrosis.

These adults include partners, parents, offspring, brothers and sisters of the family member. The aim is to find out if any of **these adults** are heterozygous for the gene that causes cystic fibrosis.

Discuss the ethical and social considerations of making a genetic test for cystic fibrosis available to these adults.

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[Total: 14]